

Programming Assignment-3

References:

1. KNN Classification Implementation: [knnClassification.ipynb](#)

Note: Submit your Python solution in a .py or .ipynb file. You can verify your solutions in Google Colab, VS Code, or any of your popular frameworks.

Application motivation: A wine seller has three different vineyards that all grow grapes in the same part of Italy. Each vineyard produces its own wine. Somewhere along the way, all the bottles got mixed up – no labels, no way to tell which wine came from which vineyard just by looking at them. But you *do* have one thing: a chemistry lab ran tests on every bottle and measured 13 things about each wine – how much alcohol it has, how acidic it is, its color, its mineral content, and so on.

The question: Using only these 13 chemical measurements, can you figure out which of the **three vineyards** each wine came from? Given a wine's chemical "fingerprint," predict **which of 3 producers made it**.

Programming Assignment Questions: Consider the **wine dataset** available in sklearn that has 178 rows and 14 columns, including the target column. Perform the KNN classification algorithm on this dataset to predict a producer class. Compute the confusion matrix and accuracy score based on your predicted results. You can use the code shared here for your reference and do necessary changes. [knnClassification.ipynb](#)

For the Iris dataset, we have used “species” as the target column name. Similarly, you can use “producer” as the target column name for the **wine** dataset.

Steps involved:

1. Load the dataset, and create a pandas DataFrame.
2. Explore the dataset and print the first 5 rows and various statistics.
3. From the dataframe, create numpy arrays X and y corresponding to feature columns and the target column.
4. Perform train-test-split-up, fit the data (training), and obtain predicted classes on test data.
5. **Print the confusion matrix and the accuracy score** based on the predicted classes and the actual classes.